AI-Driven 3D Facial Analysis: Unveiling Aesthetic & Metabolic Effects of Ketogenic Diet

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1 Introduction

The ketogenic diet (KD), rich in fats and low in carbohydrates, induces ketosis—a metabolic state that promotes fat oxidation, weight loss, and improved insulin sensitivity. While KD's systemic effects are well-documented, its influence on **facial morphology and skin metrics** remains largely unexplored. This longitudinal self-study uses the **Aura 3D Imaging System**, a patented and iF Design Award-winning technology **co-invented by the author**, to investigate facial changes alongside body composition and biochemical markers. The central hypothesis: **facial features may serve as non-invasive proxies for systemic health and aging**, bridging aesthetics and longevity through digital twin technologies.

2 Research Questions

1. How does a **ketogenic diet** influence **3D facial morphology**, **body composition**, **biochemical markers**, and **cardiovascular parameters** over a **10-week period**?
2. What **measurable correlations** exist between **facial feature changes** and **systemic health indicators**?

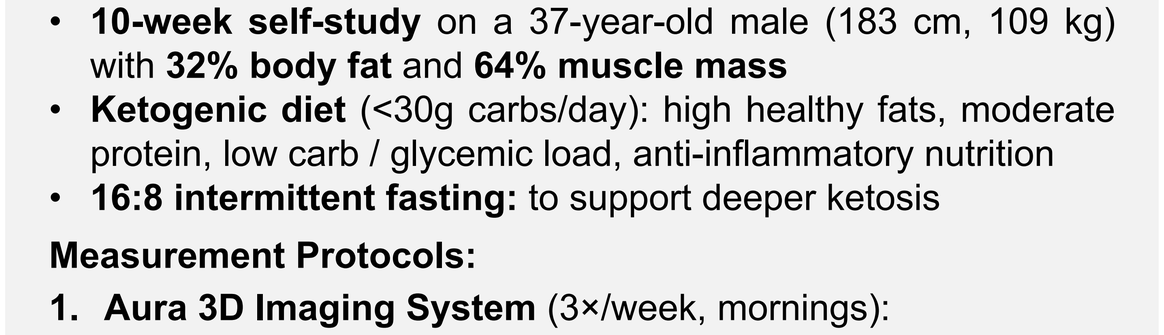
3 Methodology

Intervention Details:

- **10-week self-study** on a 37-year-old male (183 cm, 109 kg) with **32% body fat** and **64% muscle mass**
- **Ketogenic diet** (<30g carbs/day): high healthy fats, moderate protein, low carb / glycemic load, anti-inflammatory nutrition
- **16:8 intermittent fasting:** to support deeper ketosis

Measurement Protocols:

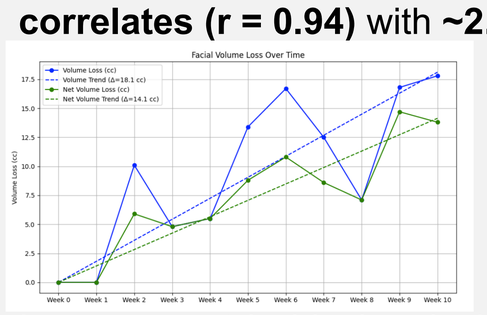
1. **Aura 3D Imaging System** (3×/week, mornings):
   0.1 mm 3D accurate face & neck analysis using photogrammetry, polarized light, and AI. Measures volume changes, wrinkles, redness, texture, pigmentation and pores
2. **Withings Body Scan** (daily, mornings):
   Segmental body composition via bioelectrical impedance analysis (fat, muscle, bone mass).
3. **Microlife BP Monitor** (daily, mornings):
   Blood pressure and heart rate tracking.
4. **Keto-Diastix** (daily, pre-meal):
   Ketone and glucose levels to monitor ketosis.
5. **Blood Tests** (pre/post intervention):
   Biochemical markers including cholesterol, ALT, testosterone, thyroid hormones, and iron.



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4 Results and Discussion

- **~5 kg weight loss (92% fat)** at ~0.5 kg/week **strongly correlates (r = 0.94)** with **~2.8 cc/kg facial volume loss**.

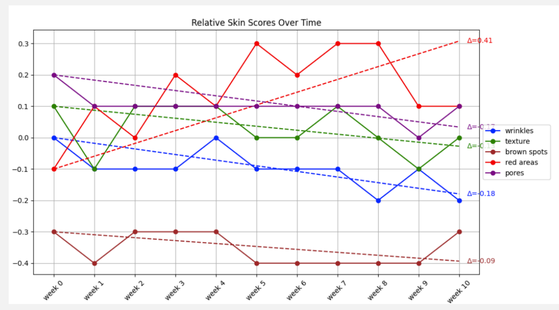




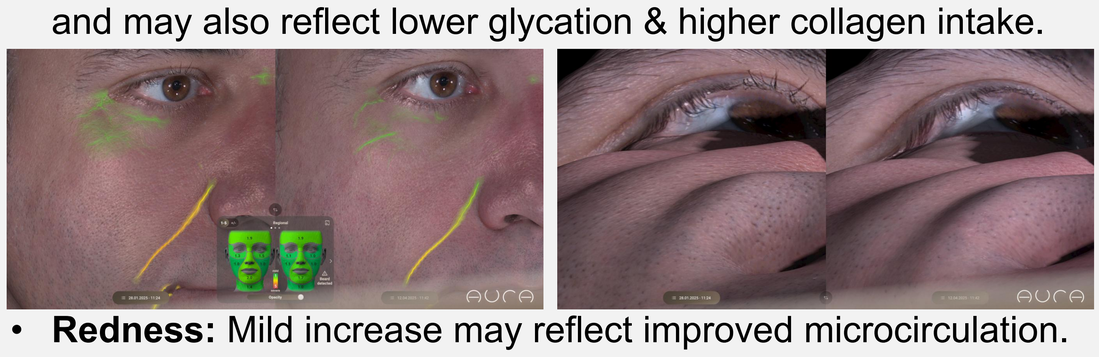
- Improvement in biochemical markers:↓Cholesterol (245→180) ↓ LDL (164→122), ↓ ALT (68→31), ↑ Testosterone (3.68→4.52)

Biochemical and Hormonal Changes

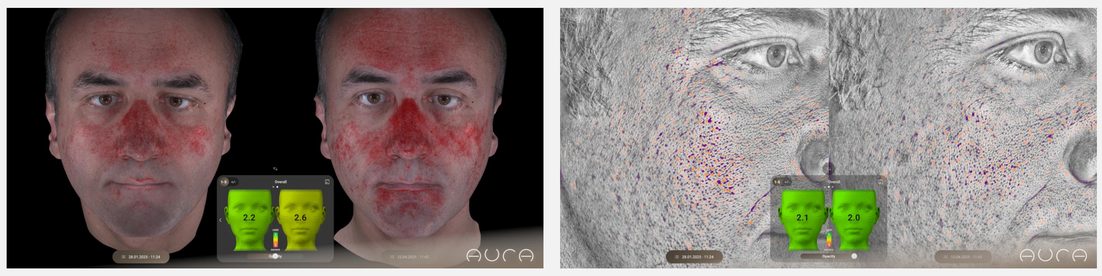
	Before Intervention: January 2025	After Intervention: April 2025	Unit	Healthy Interval
Cholesterol	245	180	mg/dL	40-200
HDL Cholesterol	52	43	mg/dL	40-60
LDL Cholesterol	164	122	mg/dL	20-100
Triglyceride	109	66	mg/dL	20-150
ALT	68	31	U/L	5-50
Creatinine	0.66	0.79	mg/dL	0.67-1.17
HGB	16.4	15.6	g/dL	12.5-16.3
HCT	50.3	46.5	%	36.2-46.3
Testosterone	3.68	4.52	µg/L	1.75-7.81
TSH	1.837	2.315	mUI/L	0.34-5.6
T3	3.83	4.19	ng/dL	2.6-4.37
T4	0.71	0.91	ng/L	0.61-1.38



- **Wrinkles:** Reduction correlates with facial fat loss (r = 0.51) and may also reflect lower glycation & higher collagen intake.



- **Redness:** Mild increase may reflect improved microcirculation.
- **Texture:** Smoother skin likely due to reduced glycation and enhanced collagen support from nutrient-rich ketogenic intake.



5 Conclusion

KD led to significant **fat loss, ~2.8 cc/kg facial volume loss, improved skin quality, and favorable biochemical changes**. Strong correlations between facial metrics and body composition suggest potential for **AI-driven 3D imaging as a non-invasive biomarker for health and aesthetic aging**, bridging aesthetics and longevity. **Future work will validate these findings in larger cohorts, develop predictive models, and explore integration into personalized longevity strategies.**

References

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5. Lemperle et al., 2001 – A classification of facial wrinkles
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